Unveiling the future: Smart University

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Hagar Mohamed – Jannah Ramadan – Rawan Sabry – Retaj Alhassan

Keywords: Lime- Gypsum –Semi Cylinder – Pyramid – Ventilation system

Abstract

The project addressed various challenges, including recycling waste for economic and environmental purposes, enhancing scientific and technological environments, adapting to climate change, and boosting Egypt's industrial and agricultural base like in these solutions, Passive Hause tree house exemplifies energy-efficient design with challenges including higher costs and climate-specific constraints. Integrating Passive Hause principles into our university project aligns with climate change mitigation and sustainability goals, offering effective solutions to reduce carbon footprint and adapt to environmental challenges, The Passive Tesla Smart House emphasizes low energy use, efficient solar power, and smart home controls for comfort and sustainability. The project aimed to develop a smart and sustainable university using renewable energy and eco-friendly materials to combat climate change and enhance environmental sustainability. The construction approach involved utilizing lime rock as a robust natural building material, with plaster and lime powder serving as adhesive materials. The university design comprised two buildings: the first featuring a cuboid first floor with strategically placed windows for thermal comfort, and a semi-cylindrical second floor with two windows for optimal lighting; the second building had a cuboid first floor with a large door and a pyramid-shaped second floor to manage light distribution. Incorporated technologies included solar panels for renewable energy, rechargeable lithium-ion batteries for energy storage, Dht22 sensors for temperature monitoring, mq-135 sensors for air quality detection, and servo motors for automated window operations. The Passive Tesla Smart House emphasizes low energy use, efficient solar power, and smart home controls for comfort and sustainability. Key features include LED lighting, solar panels with battery backup, and advanced security and water management systems. However, like the Crystal, it may face challenges with cost and operational complexity due to its advanced technologies.

introduction

When it comes to tackling the challenges faced by Egypt in dealing with the impacts of climate change and enhancing scientific and technological settings a proposal suggests the construction of a smart and sustainable university building. The goal of this initiative is to use materials and renewable energy sources effectively to address the effects of climate change. The study delves into incorporating design solutions into university architecture taking inspiration from eco friendly housing concepts such, as the Passivhaus tree house and the Passive Tesla Smart House. While these past solutions excel in energy efficiency and sustainability they also encounter obstacles related to cost and operational intricacies. For the university projects design criteria emphasis is placed on utilizing friendly materials like lime rock for construction while integrating easily maintainable renewable energy sources like solar panels and lithium ion batteries. Furthermore integrating technologies such as temperature sensors (Dht22) air quality detectors (mq 135) and servo motors for window control is seen as essential for achieving environmental conditions within the building. The proposed solution involves creating a prototype university building comprising two structures—a cuboid shaped building alongside a pyramid inspired structure—showcasing a blend of sustainability principles, with cutting edge technology. This particular design was selected to support the objective of nurturing an technological setting all the while catering to Egypts agricultural requirements using sustainable construction methods..

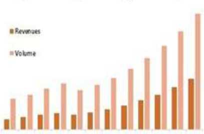

Fig (1): percentage of smart homes in Egypt

Materials and Methods

Materials	Quantity	Picture	Cost	Usage	Source
Lime	30 KG		10 pounds for each KG	Hold the bricks together	Natural Material
Arduino uno R3	1		450 Pounds	Programs can be loaded on to it from the easy-to-use Arduino computer program.	Electronics store
Micro servo motor	3		85 Pounds	Control the movement of doors and windows	Electronics store
Air quality sensor MQ135	1		135 Pounds	Monitor the concentration of the toxic gases.	Electronics store
Solar panels 4 watt	2		400 pounds	Source of energy in the building.	Electronics store
Temperature Sensor DHT 11	1		200	Measure the temperature inside the building	Electronics store
Rechargeable lithium battery	3		60 pound	Store the energy that comes from the solar panels	Electronics store
gypsum	30 KG		10 Pound for each kg	Hold the bricks together	Natural material
1s TP4056 1A	1		50 pound	Lithium battery charging module	Electronics store
Flame sensor	1		50 pounds	Table 1 (materials) detect heat, smoke, and fire.	Electronics store

Table 1 (materials)

Method:.

We used limestone rocks and lime material, and natural gypsum as a building material. There are the steps that we did in building the prototype:. First, we broke the limestone rocks and made small brick shapes to build the building. Second, We used a mixture of lime and gypsum to stick the bricks together. Third , We started building our structure with large cuboid and left four openings for windows. Fourth ,we built smaller cuboid in front of the larger one and left space for a door. Fifth , We built a semi cylinder using a mold, placed it on top of the large cuboid. Sixth , we built a pyramid on a mold as well, and placed it in front of the smaller cuboid. Finally, we left the prototype to dry. the smart system, we did as the following:. We programmed the temperature sensor to measure the temperature and send a message with the temperature it measured by using a Bluetooth module, and we made its connections to the board. We programmed the air quality sensor to measure gases like Ammonia (NH_3), sulfur (S), Benzene (C_6H_6), CO_2,.We programmed the flame sensor to detect smokeless liquid and smoke that can create open. We used 6 servo motors for each window and programmed it to open the windows which is made of glass and its in the event of a fire when the temperature measured by the sensor was more than 22, and we made its connections to the board. We used the solar panel to charge the batteries we use via the SIM 1s TP4056 1A. Therefore, our entire system is running with clean energy, which is the solar panel.

Fig (2) building

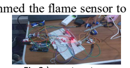

Fig 3)smart system

Test plan:.| Test plan After finishing building our prototype, we wanted to test the prototype to see if it meets the design requirements so we decided to test the prototype by the following process: First: we will measure the temperature inside the building using the temperature sensor in our smart system. We do this to see if the temperature inside the building is 3 degrees less than the temperature in shadow to test if the design of the building and thermal comfort has reduced the temperature inside the building. Second: we will test the smart system by the following way: We will test the temperature sensor and air quality sensor by uploading the code and see if the sensor is working by checking the readings. We will test the flame sensor by uploading the code and bring a fire source and see the readings of the sensor. We will test the servo motors by increasing the temperature inside the building or bring a flame source near to flame sensor, because we these are the conditions that will make the servo motors move.

Results

After testing our prototype we came up with many results:
first we tested the temperature by temperature sensor and results that we got as the following:

Temperature	Under the sun	In shadow	Our result
Degree	28 ‘C	26’c	23’c

Table2 (result)

Another result to our project that the windows open when the degree of temperature become more than 22’c and that happened when we made our test as the degree of temperature of the building was 23’c so, the windows opened.
flame sensor detect the fire so, we programmed that when there is any fire in the building , this sensor will send notification to the mobile phone and we tested that by that we brought a lighter near to the flame sensor, so the flame sensor activated and gave us a message to our mobile phone.

Analysis

as a passive house which is ecofriendly and minimizes the impact on the environment and is built with eco-friendly material. We construct the prototype in many steps that achieve the design requirement and solve these grand challenges. First, we cut the lime rock into bricks to make the main building bricks. Then, we made a mixture of equal amounts of lime and gypsum with water as a paste to stick bricks Egypt faces many challenges. Climate change is from grand challenge which Egypt faces example transportation and industry due to burning fossil fuels to get the energy to work in transportation and industry leads to an increase in greenhouse gases and toxic gases like burning coal, oil, or gas that lead to increased levels of carbon and nitrous oxide lead to bad impact on the environment as increasing temperature because the high level of carbon avoids heat to come back out of the atmosphere that leads to increase drought and low level of water because of evaporate them. Another reason for climate change is deforestation for creating farms or factories leads to increased carbon also and the same negative feedback. Another challenge is pollution and recycling garbage. The prior solutions to these challenges are the Tesla house, the Passivhaus tree house, and eco-friendly domes for living. Passive Hause is known together. First, the back section consists of a large cuboid with a semi-cylinder from the top Second, the front section consists of a small cuboid with a pyramid from the top. In addition to four windows in a large cuboid (two are in the back and two in the sides), two sides of a semi-cylinder and a big door in the front view. Third, we used molds of shapes semi-cylindrical and one pyramid with dimensions we needed from the lime and gypsum paste until it took the mold shape. Then we stick them to other parts of the prototype with the same mixture of paste. Finally, the shape of the prototype is separated into main sections, and all of the above is about the section of building design, the second section is about the smart system and the technology footprint that are made by using: solar panels as a source of renewable and clean energy. rechargeable lithium Ion Battery for storing energy that is done by solar panels. servo motors for controlling the moving of the windows of the university. Dht11 sensor to measure the degree of temperature inside the university. flame sensor to detect heat, smoke, and fire inside the building. mq135 sensor to Monitor the concentration of the toxic gases. Arduino Uno R3 for controlling and doing all operations of the digital components. And to make sure that all operations will go well and continue, the energy consumption should be calculated.\calculating the energy consumption of the system for an active hour and a half: Total power consumption = Solar panels + Batteries + Servos + DHT11+ Flame sensor + MQ135 + Arduino Uno R3
= 8W + 108Wh + 7.056W + 0.0125W + 0.035W + 0.75W + 0.25W ≈ 115.1035 watts. Given the power consumption, we can calculate the energy consumption for the active hour and a half: Energy consumption = Total power consumption * Time = 115.1035 watts * 1.5 hours ≈ 172.65525 watt-hours.
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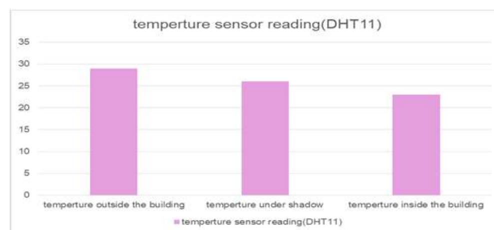

Graph(1)

Another result to our project that the windows open when the degree of temperature become more than 22’c and that happened when we made our test as the degree of temperature of the building was 23’c so, the windows opened.
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<u>In Chemistry LO:09:</u> The path of electricity in the battery: In the battery, electrons move from anode to cathode through an external storage system which makes us understand the importance of this path to design effective energy storage systems by using batteries in the smart building.
<u>Mechanics LO:05:</u> Power: It generates renewable energy sources such as solar, wind, and hydropower is important for sustainable building. Using these sources the design of smart building reduces dependence on non-renewable energy and keeps the sustainability of the environment.
<u>Physics LO:10:</u> Calculating the effective value of EMF ensures the efficient use of electrical power, reducing energy wastage and promoting sustainable practices.

conclusion

In conclusion, concerning the grand challenges facing Egypt such as climate change adaptation and sustainable development, this essay proposed the prototype of a smart, sustainable university building. This project has been carried out utilizing eco-friendly material like lime rock and renewable energy resources such as solar panels and lithium-ion batteries. Key takeaways from the discussion include the feasibility of implementing passive design strategies and modern technologies to improve the sustainability of the environment. The conclusion based on the discussion from the essay reiterates the possibility of building eco-friendly university architecture based on industrial and agricultural needs in Egypt to counter climate change.

recommendation

Small scale (prototype)
1- Increase the number of Air suction device to increase Ventilation .
2- The windows are made of two panels of glass with air between them.
3- Make the solar panel move at a specific angle at a specific time.
large scale:
1-Increasing the number of floors of the building and the rooms.
2- Increasing the number of windows in the building.
3- Make a fountain to reduce the temperature.
4-making two foundations (bases) at the building for the university , the first one with parties of an angle 90 , and the second one with parties of an angle less than 90 , this process should be done to increase stability and break the thermal bridges at the university.
5-In the winter, the ventilation system will not depend on windows, but rather on a hood.

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For Further information

Hagar Mohamed:hagar.1722513@stemismailia.moe.edu.eg
Jannah Ramadan:Jana.1722524@stemismailia.moe.edu.eg
Rawan Sabry:Rawan.1722546@stemismaillia.moe.edu.eg
Retaj Alhassan:Retaj.1722547@stemismailia.moe.edu.eg